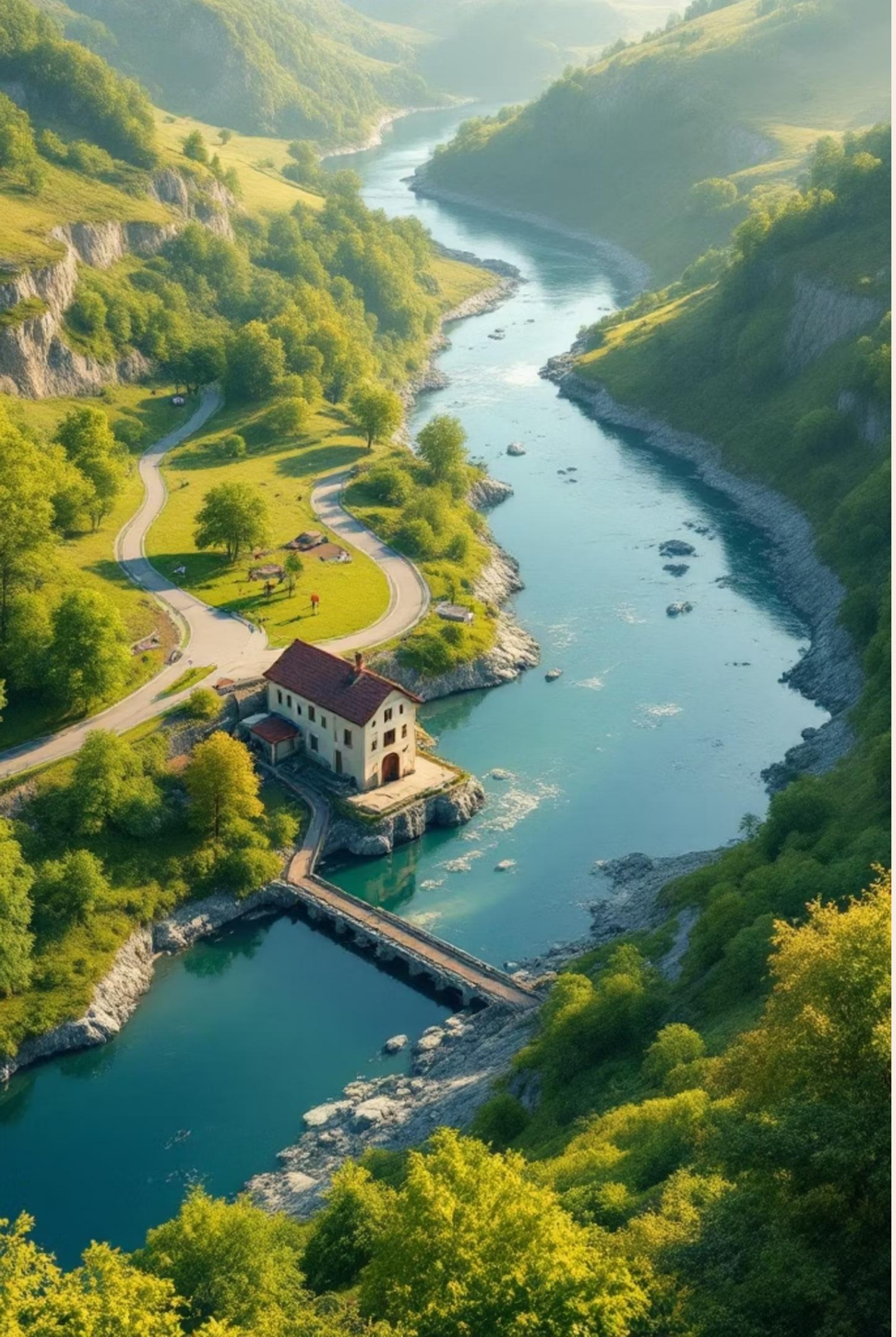
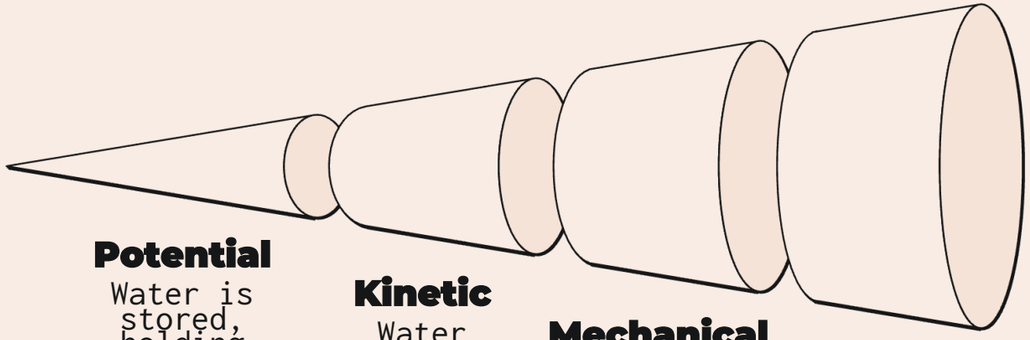


Small Hydropower for Rural Electrification

Hydropower systems up to 10 MW that harness flowing water from rivers, streams, or canals to produce electricity. Ideal for remote and rural communities — clean, reliable, and one of the oldest widely used renewable technologies.

- Capacity range: micro (up to 100 kW), mini (100 kW–1 MW), small (1–10 MW). This deck focuses on practical design, types, and deployment considerations for planners and engineers.

How Small Hydropower Works



Potential
Water is stored, holding potential energy.

Kinetic
Water flows, creating kinetic energy.

Mechanical
The turbine shaft spins from water force.

Electrical
Energy is converted into usable electricity.

Water is diverted or stored, channeled through a penstock to accelerate flow and strike turbine blades. The turbine turns a shaft coupled to a generator, converting mechanical rotation into electrical power.

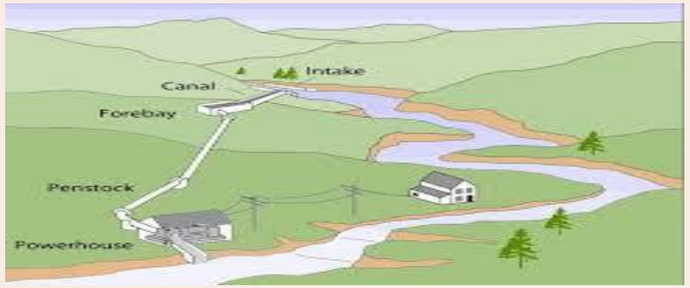
Key Components

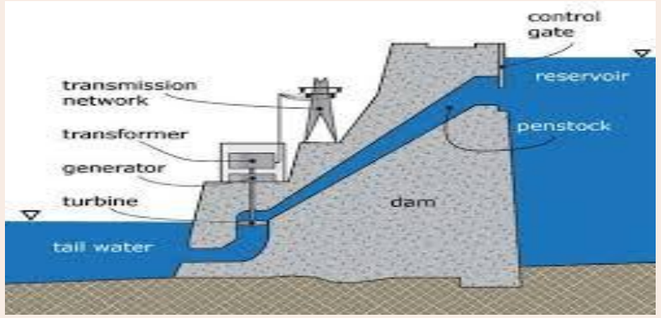
- Intake & diversion structure
- Penstock (pressurized conduit)
- Turbine & governor
- Generator & control systems

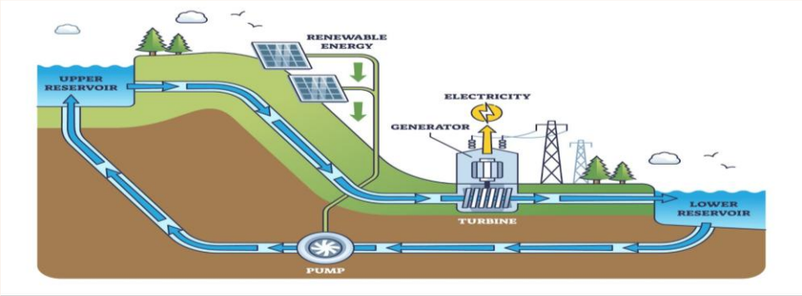
Energy Conversion

Potential → Kinetic → Mechanical → Electrical. Head and flow determine available power ($P \approx \rho g Q H \eta$).

Types of Small Hydropower







Run-of-River

Water diverted directly from a flowing stream. No large reservoir — lower footprint and faster permitting. Best where flow is steady.

Reservoir-Based

Small dam stores water for regulated generation — offers dispatchability and buffering for dry periods but larger civil works required.

Pumped Storage (Small-scale)

Water is pumped to an upper reservoir during low-demand periods and released to generate when needed. Rare at small scale but useful for local storage.

These are scaled-down approaches of conventional hydroplants — choose by site constraints: head, flow variability, land use, and community needs.

Advantages & Limitations — Practical Lens

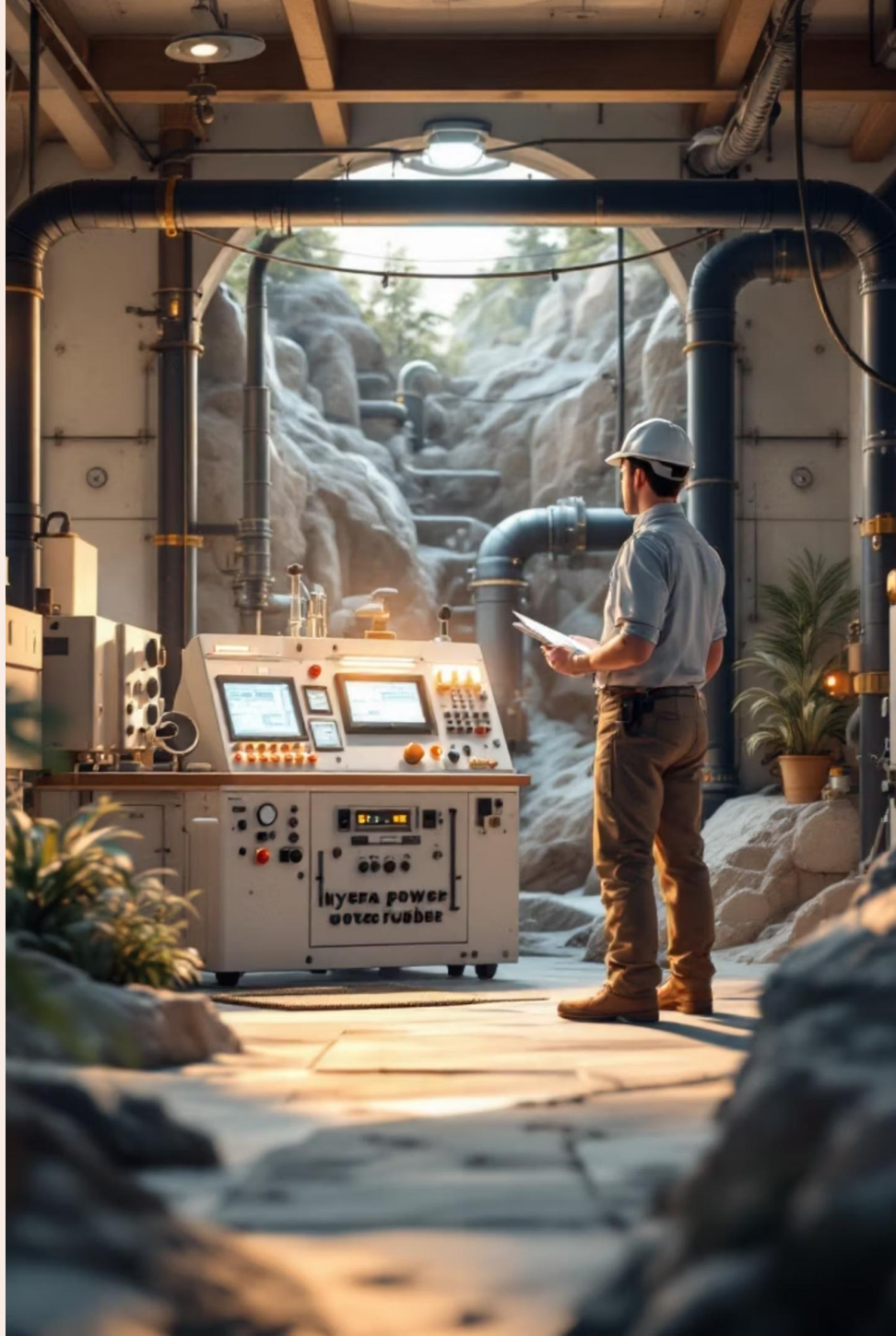
Advantages

- Renewable, low-emissions baseload where flow is steady
- Lower ecological and social impacts vs large dams
- Enables decentralized electrification — villages, clinics, schools
- Long asset life with modest O&M

Limitations

- Seasonal and climate-sensitive — dry seasons reduce output
- Site-specific: requires sufficient head and reliable flow
- Higher upfront civil and permitting costs per kW
- Ecological concerns: fish passage, sediment, water rights

Mitigation strategies: hydraulic modeling, small fish-friendly intakes, seasonal complementarity with solar/battery, community engagement, and environmental flow allocations.



Applications, Deployment Steps & Next Actions

1. Village Electrification

Provides reliable electricity for homes, street lighting, and basic community needs in remote areas.

2. Agro-Processing Units

Powers rice mills, flour mills, oil extraction units, and irrigation pumps — supporting local agriculture.

3. Small & Rural Industries

Supplies energy to workshops, textile units, carpentry shops, and small manufacturing businesses.

4. Health Centers & Telecom Sites

Ensures continuous power for medical equipment, refrigeration (vaccines), communication towers, and emergency services.

Conclusion:

Small hydropower plants provide a clean, reliable, and decentralized energy solution for rural and remote regions. With proper site selection, community involvement, and sustainable design, they can deliver long-term electricity with minimal environmental impact.

By integrating with solar and battery systems, small hydropower becomes even more resilient — making it a practical and sustainable option for meeting local energy needs and supporting economic development.

THANK YOU